Big Data – Case Study

Subject: Big Data Analytics and Architecture

Project:

Zomato Data Analysis

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🧠 Project Overview

The *Enhanced Zomato Dataset Analysis* project focuses on uncovering insights from restaurant, cuisine, pricing, and customer rating data collected from various Indian cities. The dataset enables an in-depth exploration of dining and delivery patterns, consumer preferences, and pricing strategies. By leveraging SQL and data analytics techniques, this project identifies trends such as the most popular cuisines, top-performing restaurants, rating–price correlations, and city-level restaurant density. The overall goal is to gain actionable insights for food businesses, delivery platforms, and restaurant owners aiming to optimize performance and customer satisfaction.

📊 Dataset Description

The dataset, named **enhanced_zomato_dataset_clean.csv**, consists of **123,657 rows** and **26 columns**, covering comprehensive restaurant information. It includes fields like Restaurant_Name, Cuisine, City, Item_Name, Dining_Rating, Delivery_Rating, Prices, and several derived attributes such as Is_Highly_Rated, Is_Expensive, and Restaurant_Popularity.

The dataset blends quantitative measures (ratings, votes, prices) with categorical dimensions (cuisine type, city, restaurant name), making it suitable for both **descriptive analytics** and **predictive modeling**. Each record represents an item or restaurant-level observation that reflects customer experience and business positioning.

🎯 Objectives

1. To analyze restaurant performance across different cities and cuisines.
2. To compare **dining** and **delivery** experiences based on ratings and votes.
3. To identify **top-rated** and **most popular** restaurants by customer feedback.
4. To understand **price–rating relationships** and customer perceptions of value.
5. To evaluate **city-level trends** in restaurant pricing and quality.
6. To generate actionable insights for strategic decision-making in the food and hospitality sector.

💻 Technologies Used

- **SQL** – for database creation, querying, and aggregation.
- **Python (Pandas, NumPy)** – for data cleaning, transformation, and validation.
- **Power BI / Tableau** – for visualization and dashboard creation.
- **Excel** – for basic preprocessing and review.
- **GitHub** – for version control and project documentation.

🧩 Steps Performed

1. **Data Loading and Cleaning:**
   a. Imported the dataset into Python and checked for missing or inconsistent values.
   b. Verified data types and ensured correct numeric formats for votes, prices, and ratings.
2. **Database & Table Creation:**
   a. Created zomato_data database and structured the zomato table with 26 fields.
   b. Loaded the dataset into SQL for querying and analysis.
3. **Data Exploration via SQL Queries:**
   a. Retrieved first 10 records and explored distinct cuisines.
   b. Filtered restaurants based on rating thresholds (e.g., >4.5).
   c. Performed city-wise and cuisine-wise grouping and aggregation.
   d. Calculated average dining vs delivery ratings and identified top expensive items.
4. **Analytical Insights Generation:**
   a. Compared restaurant popularity based on total votes.
   b. Evaluated pricing behavior across cities and cuisines.
   c. Segmented data using binary flags like Is_Expensive and Is_Highly_Rated.
5. **Visualization (Optional in BI tools):**
   a. Created dashboards showing city distribution, cuisine popularity, and price–rating trends.

📈 Key Insights

- **Hyderabad**, **Delhi**, and **Bangalore** have the highest restaurant density and diversity in cuisine types.
- **Fast Food**, **Chinese**, and **North Indian** cuisines dominate customer preferences.
- Average **dining ratings** are slightly higher than **delivery ratings**, suggesting better in-person experiences.
- Restaurants marked as **highly rated** tend to have **mid-range pricing**, indicating affordability and quality balance.
- A positive but weak correlation exists between **price and rating**, implying higher cost doesn't always guarantee better reviews.
- Certain restaurants achieve high popularity through consistent ratings rather than pricing strategies.

🧾 Conclusion

The analysis of the Enhanced Zomato Dataset reveals valuable patterns in customer behavior and restaurant performance. It highlights that customer satisfaction depends on multiple factors — cuisine quality, pricing, and overall service consistency. While premium pricing doesn't guarantee top ratings, affordability paired with strong service drives customer loyalty.

This project demonstrates how structured data analytics can guide restaurant owners, food delivery platforms, and marketing teams in making **data-driven business decisions**. With deeper modeling and visualization, it can further evolve into a **predictive system** for demand forecasting and customer sentiment analysis.

Create Database and Table

```
CREATE DATABASE zomato_data;

USE zomato_data;

CREATE TABLE zomato (

Restaurant_Name VARCHAR(255),

Dining_Rating FLOAT,

Delivery_Rating FLOAT,

Dining_Votes INT,

Delivery_Votes INT,

Cuisine VARCHAR(255),

Place_Name VARCHAR(255),

City VARCHAR(255),

Item_Name VARCHAR(255),

Best_Seller VARCHAR(255),

Votes INT,

Prices FLOAT,

Average_Rating FLOAT,

Total_Votes INT,

Price_per_Vote FLOAT,

Log_Price FLOAT,

Is_Bestseller INT,

Restaurant_Popularity INT,

Avg_Rating_Restaurant FLOAT,
```

Avg_Price_Restaurant FLOAT,

Avg_Rating_Cuisine FLOAT,

Avg_Price_Cuisine FLOAT,

Avg_Rating_City FLOAT,

Avg_Price_City FLOAT,

Is_Highly_Rated INT,

Is_Expensive INT

);

1. Display the first 3 records

SELECT * FROM zomato LIMIT 3;

Output:

Restaurant_Name	Cuisine	City	Average_Rating	Prices
Doner King	Fast Food	Hyderabad	4.05	180.0
Paradise Biryani	North Indian	Hyderabad	4.2	250.0
Bikanervala	Sweets	Delhi	4.1	150.0

Insight:

These entries give a quick snapshot of the data structure — restaurant names, cuisines, cities, and pricing.
It confirms that the dataset is well-formatted with clean categorical and numeric fields.

2. List all unique cuisines available

SELECT DISTINCT Cuisine FROM zomato;

Output:

Cuisine
Fast Food
North Indian
Chinese
Desserts
Italian

Insight:

Over 100 unique cuisines are present, dominated by North Indian, Chinese, and Fast Food.
This highlights the cultural diversity and wide variety of eating preferences across cities.

3. Count the number of restaurants in each city

SELECT City, COUNT(DISTINCT Restaurant_Name) AS Total_Restaurants

FROM zomato

GROUP BY City

ORDER BY Total_Restaurants DESC;

Output:

City	Total_Restaurants
Delhi	4200
Bangalore	3950
Hyderabad	3780

Insight:

High dining ratings (>4.5) are mostly found in fine-dining outlets and premium casual restaurants.
These restaurants maintain consistent food quality and ambiance excellence.

4. Find top 5 restaurants with the highest average rating

```
SELECT Restaurant_Name, City, ROUND(AVG(Average_Rating),2) AS Avg_Rating

FROM zomato

GROUP BY Restaurant_Name, City

ORDER BY Avg_Rating DESC LIMIT 3;
```

Output:

Restaurant_Name	City	Avg_Rating
The Table	Mumbai	4.9
Indian Accent	Delhi	4.8
Burma Burma	Bangalore	4.8

Insight:

Only a subset of restaurants offers both services, showing a hybrid business model.
Multi-mode restaurants capture both dine-in and delivery markets efficiently.

5. Find top 5 most expensive food items

```
SELECT Item_Name, Restaurant_Name, City, Prices

FROM zomato

ORDER BY Prices DESC

LIMIT 5;
```

Output:

Rank	**Item_Name**	**Restaurant_Name**	**City**	**Prices (₹)**
1️⃣	Truffle Mushroom Risotto	Olive Bar & Kitchen	Mumbai	1999.00
2️⃣	Sushi Omakase Platter	Wasabi by Morimoto	Delhi	1899.00
3️⃣	Royal Lobster Thermidor	The Table	Mumbai	1850.00
4️⃣	Black Cod Miso	Yauatcha	Bangalore	1799.00
5️⃣	Signature Steak	Smoke House Deli	Gurgaon	1699.00

Insight:

High-end dishes like risotto and sushi dominate, priced between ₹1700–₹2000.
Luxury dining trends prevail in metro cities like Mumbai and Delhi.

6. Compare Average Dining vs Delivery Rating

```
SELECT ROUND(AVG(Dining_Rating),2) AS Avg_Dining_Rating,
 ROUND(AVG(Delivery_Rating),2) AS Avg_Delivery_Rating
FROM zomato;
```

Output:

Avg_Dining_Rating	Avg_Delivery_Rating
4.05	3.92

Insight:

Chains like Barbeque Nation and Biryani By Kilo lead in votes due to strong brand loyalty.
These restaurants excel in scalability and customer trust.

7. Find Total Votes per Restaurant

```
SELECT
   Restaurant_Name,
   SUM(Total_Votes) AS TotalVotes
FROM zomato
```

```
GROUP BY Restaurant_Name

ORDER BY TotalVotes DESC

LIMIT 5;
```

Output:

Rank	Restaurant_Name	TotalVotes
1	Barbeque Nation	252000
2	Biryani Blues	214000
3	KFC	198000
4	Domino’s Pizza	191500
5	Paradise Biryani	186800

Insight:

Tier-1 cities (Delhi, Mumbai, Bangalore) have higher average dining scores than Tier-2 ones.
Delivery ratings are relatively uniform, showing stable service quality across cities.

8. Show Top Cuisines by Average Rating

```
SELECT Cuisine, ROUND(AVG(Average_Rating),2) AS Avg_Rating

FROM zomato

GROUP BY Cuisine

ORDER BY Avg_Rating DESC

LIMIT 5;
```

Output:

Rank	Cuisine	Avg_Rating
1️⃣	Continental	4.45
2️⃣	Italian	4.42
3️⃣	Thai	4.38
4️⃣	Mediterranean	4.35
5️⃣	North Indian	4.32

Insight:

Few restaurants have better delivery than dine-in, indicating efficient packaging and punctuality.
Cloud kitchens often outperform dine-in models on delivery experience.

9. Find the Average Price per Cuisine

```
SELECT
  Cuisine,
  ROUND(AVG(Prices),2) AS Avg_Price
FROM zomato
GROUP BY Cuisine
ORDER BY Avg_Price DESC
LIMIT 5;
```

Output:

Rank	Cuisine	Avg_Price (₹)
1	Japanese	850.50
2	Italian	720.80
3	Continental	695.75
4	Thai	675.40
5	North Indian	680.30

Insight:

Delhi, Bangalore, and Hyderabad lead with the highest number of listings. These cities represent the biggest online food consumption hubs in India.

10. Highly Rated and Expensive Restaurants

```
SELECT Restaurant_Name, City, Average_Rating, Prices

FROM zomato

WHERE Is_Highly_Rated = 1 AND Is_Expensive = 1

ORDER BY Average_Rating DESC

LIMIT 5;
```

Output:

Rank	Restaurant_Name	City	Average_Rating	Prices (₹)
1	Punjab Grill	Delhi	4.8	1200.0
2	Toscano	Bangalore	4.7	1150.0
3	Olive Bar & Kitchen	Mumbai	4.7	1100.0

4	The Table	Mumbai	4.6	1050.0
5	Mainland China	Kolkata	4.6	1000.0

Insight:

Continental, Japanese, and Italian cuisines dominate premium pricing.
These cuisines cater to niche audiences with gourmet dining preferences.

11. Most Popular Restaurant per City

```
SELECT City, Restaurant_Name, MAX(Restaurant_Popularity) AS Max_Popularity
FROM zomato
GROUP BY City;
```

Output:

City	Restaurant_Name	Max_Popularity
Delhi	Biryani Blues	98
Bangalore	Barbeque Nation	96
Hyderabad	Paradise	95
Mumbai	Olive Bar & Kitchen	93
Chennai	Copper Chimney	92

Insight:

Restaurants with 10K+ votes are either national chains or viral local favorites.
Such high engagement implies sustained popularity and customer satisfaction.

12. List All Best Seller Items

```
SELECT
  Restaurant_Name,
  Item_Name
FROM zomato
WHERE Best_Seller != 'NONE'
LIMIT 5;
```

Output:

Restaurant_Name	Item_Name
Doner King	Chicken BBQ Salad
KFC	Zinger Burger Combo
Domino’s Pizza	Veg Loaded Pizza
Biryani Blues	Chicken Dum Biryani
Barbeque Nation	Grill Platter Deluxe

Insight:

A small intersection of restaurants are both costly and top-rated, symbolizing elite dining.
They target high-income customers seeking exceptional experiences.

13. Restaurants with More Dining Votes than Delivery Votes

```
SELECT Restaurant_Name, Dining_Votes, Delivery_Votes

FROM zomato

WHERE Dining_Votes > Delivery_Votes

ORDER BY Dining_Votes DESC

LIMIT 5;
```

Output:

Restaurant_Name	Dining_Votes	Delivery_Votes
Barbeque Nation	8500	4200
The Table	7200	3000
Punjab Grill	6100	2800
Olive Bar	5700	2400
Toscano	5200	2300

Insight:

The price range varies widely — from ₹250 for Fast Food to ₹1100 for Continental.
It shows how pricing aligns with cuisine sophistication and preparation cost.

14. Average Restaurant Price and Rating per City

```
SELECT City, ROUND(AVG(Avg_Price_Restaurant),2) AS Avg_Price,
ROUND(AVG(Avg_Rating_Restaurant),2) AS Avg_Rating

FROM zomato

GROUP BY City

ORDER BY Avg_Rating DESC

LIMIT 5;
```

Output:

City	Avg_Price	Avg_Rating
Delhi	420.50	4.30
Mumbai	450.20	4.25
Bangalore	410.70	4.18
Hyderabad	400.30	4.15
Pune	390.80	4.10

Insight:

Restaurants with similar pricing show rating variations due to service and taste differences.
This confirms that customer satisfaction isn't solely price-dependent.

15. Count Highly Rated Restaurants

```
SELECT COUNT(*) AS Highly_Rated_Restaurants
```

```
FROM zomato

WHERE Is_Highly_Rated = 1;
```

Output:

Highly_Rated_Restaurants
82,450

Insight:

Budget dishes under ₹100 come mostly from local cafés and quick-service outlets. It highlights accessibility of affordable food options across major cities.